Web环境搭建（新）

下载配置Chrome和ChromeDriver

下载配置Maven

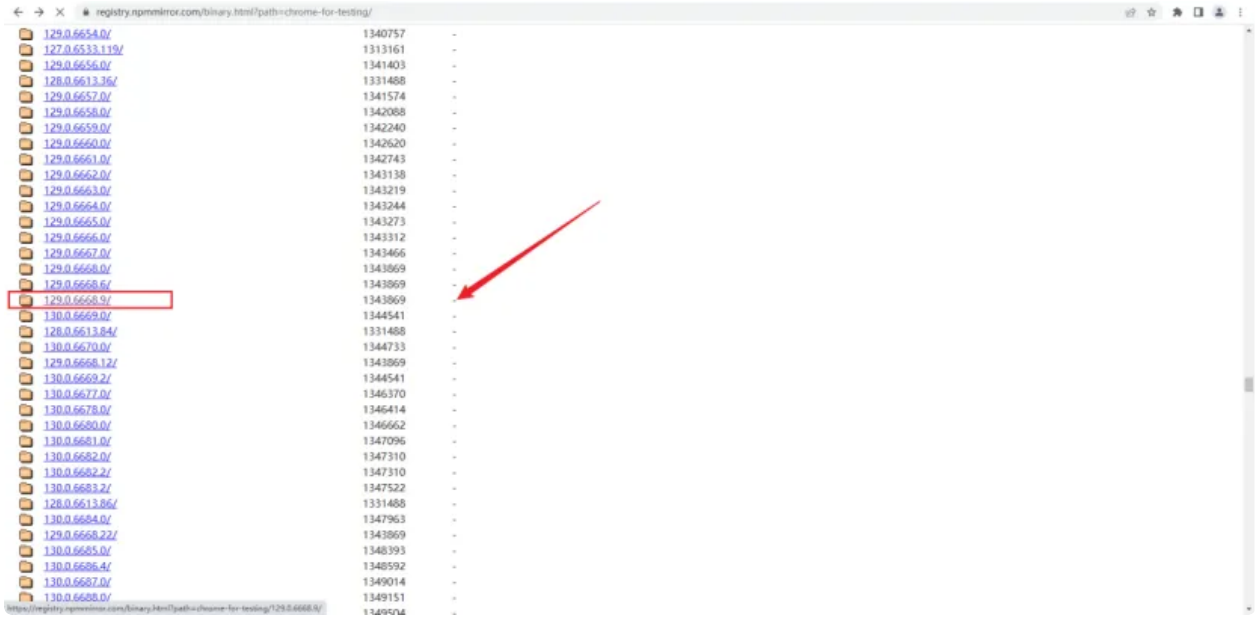
IDEA创建Maven项目

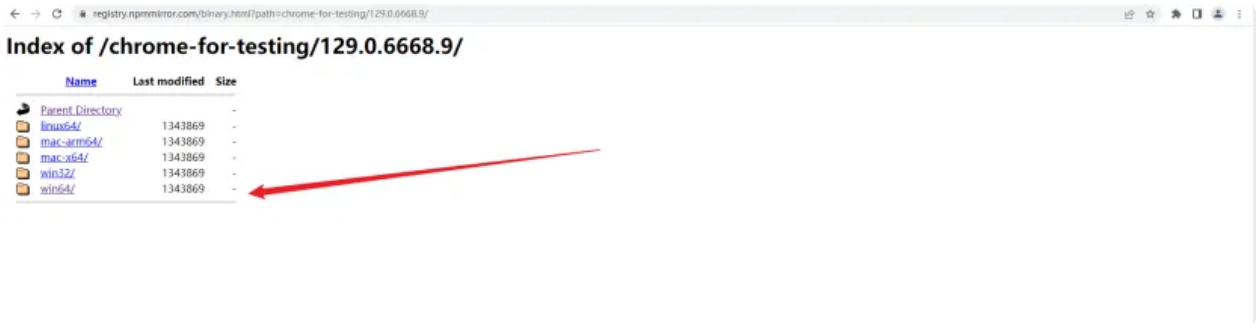
下载配置Chrome和ChromeDriver

1. 下载谷歌浏览器(此处安装步骤不在此描述)
2. 下载配置ChromeDriver
   a. 首先我们要去看自己的谷歌浏览器版本,如下图操作



   b.
   c. 接着我们去下载对应版本的驱动，在下面这个网址中查找对应版本的安装包并下载谷歌浏览器驱动。https://registry.npmmirror.com/binary.html?path=chrome-for-testing/

d. 

e. 

f. 

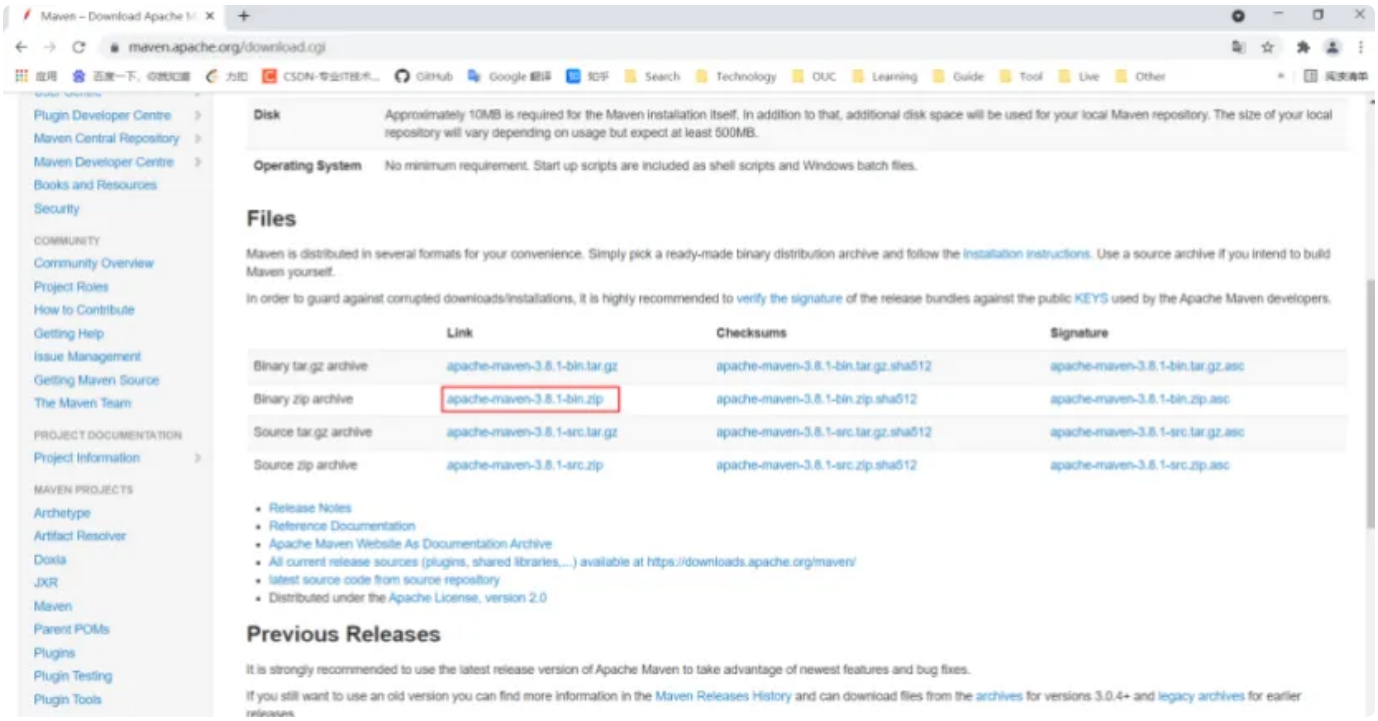
g. 下载完后解压并保存一下运行程序的地址，后面会用到

h. 

下载配置Maven

一、Maven (项目管理工具)安装

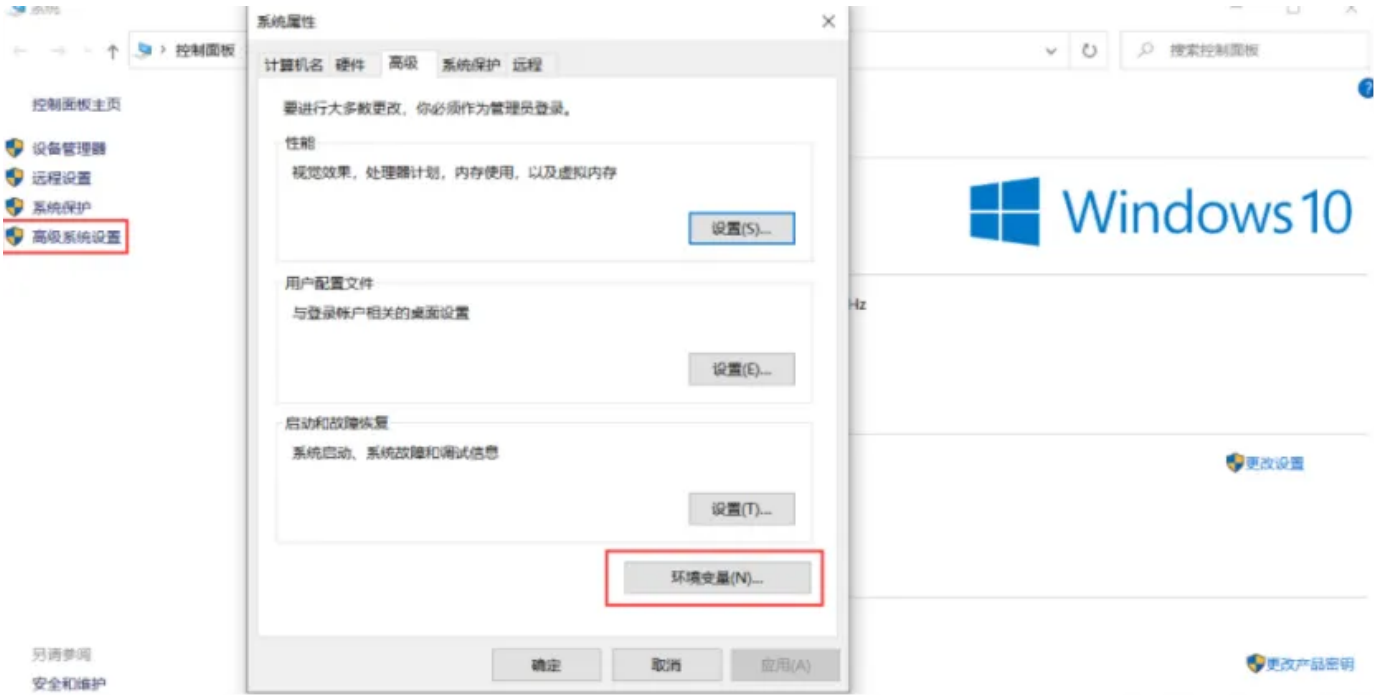
1. 官网下载（https://maven.apache.org/download.cgi）



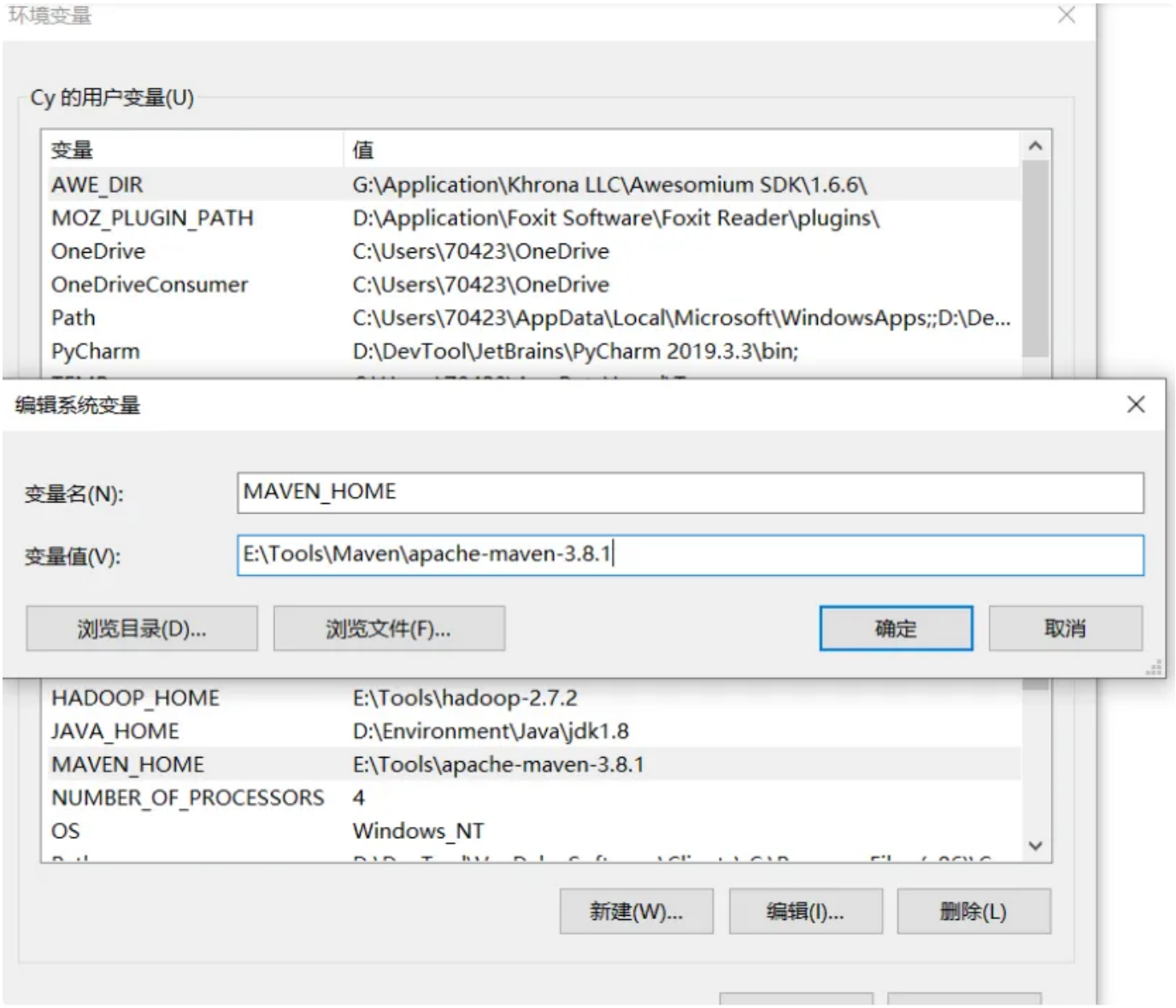
2. 下载完成后，解压到某一路径下。本文以E:\Tools\Maven\apache-maven-3.8.1为例，实际配置环境变量时以自己安装的路径为准。

二、配置环境变量

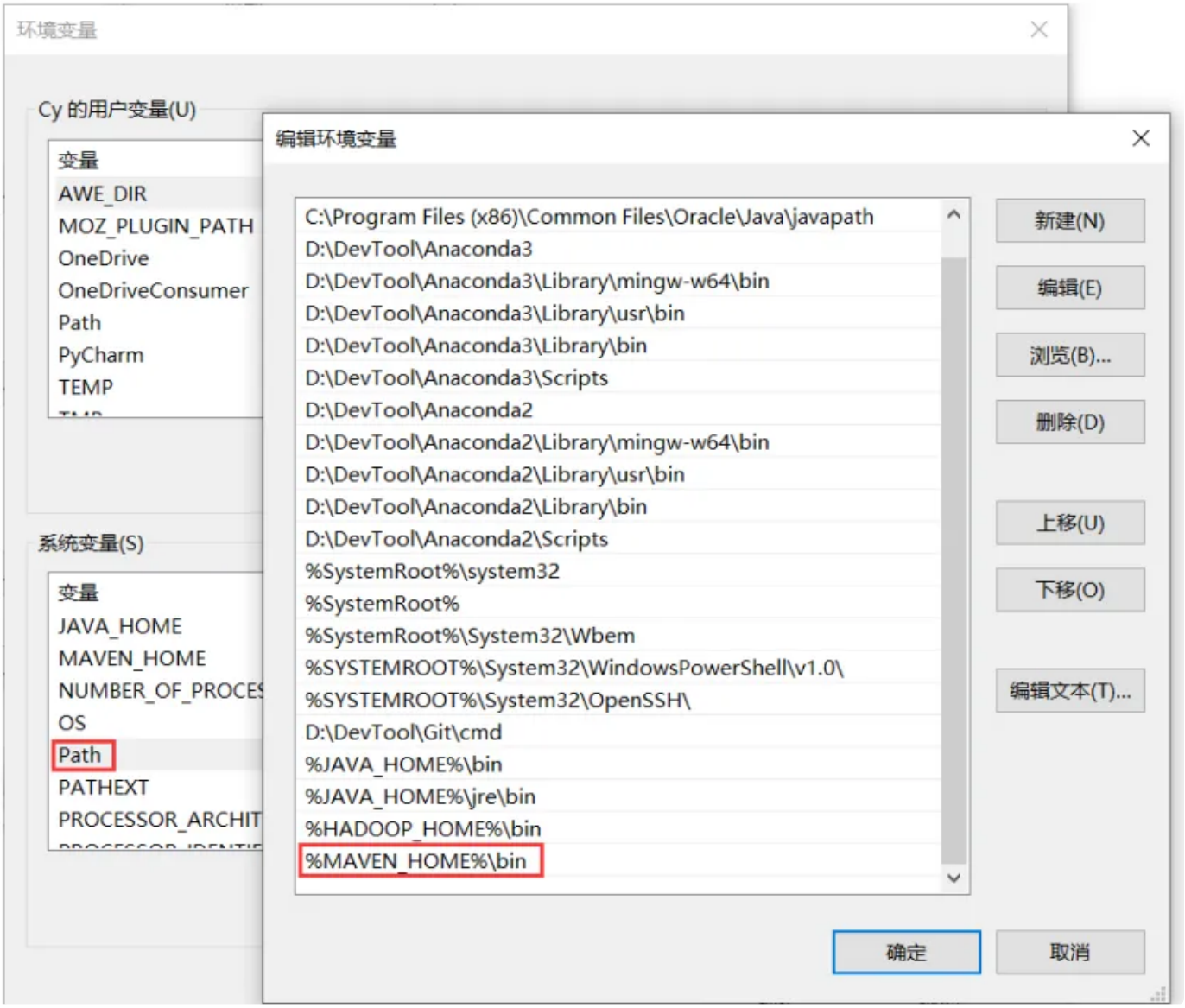
1.右键此电脑->属性->高级系统设置->环境变量



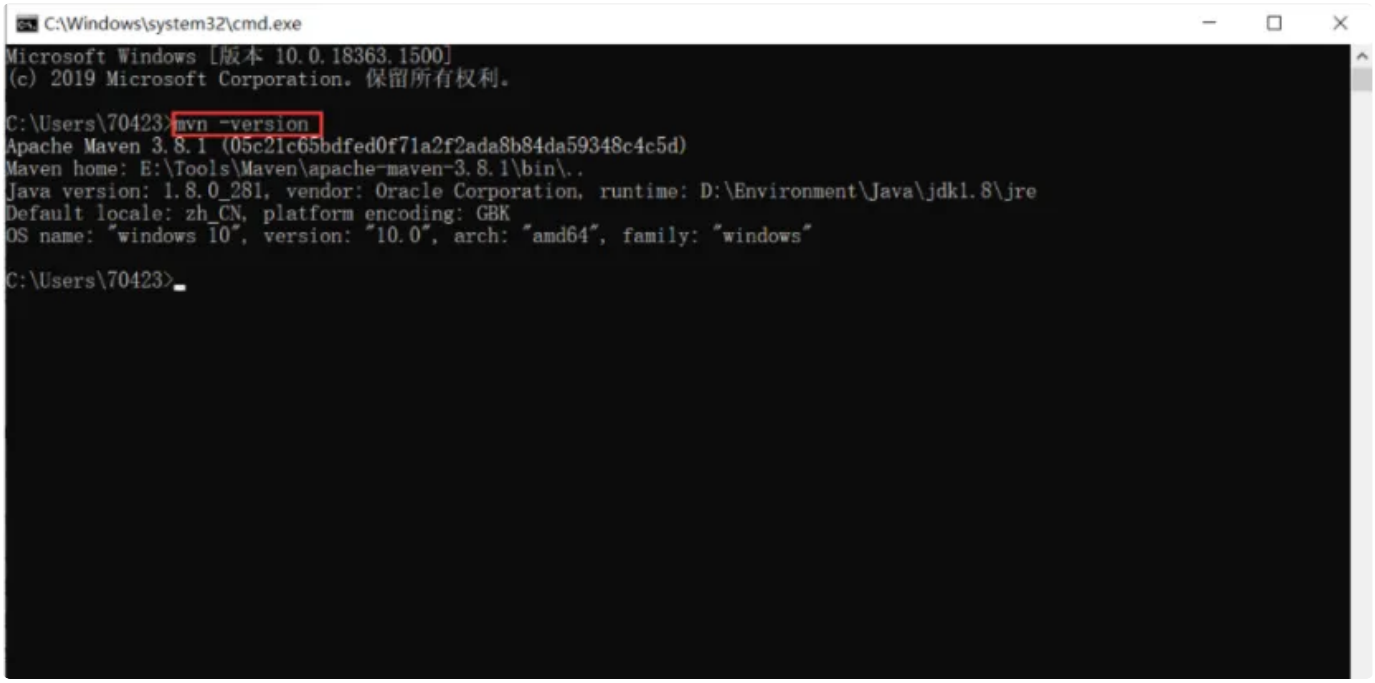
2.新建变量MAVEN_HOME = E:\Tools\Maven\apache-maven-3.8.1（以自己的安装路径为准）



3.编辑变量Path，添加变量值%MAVEN_HOME%\bin

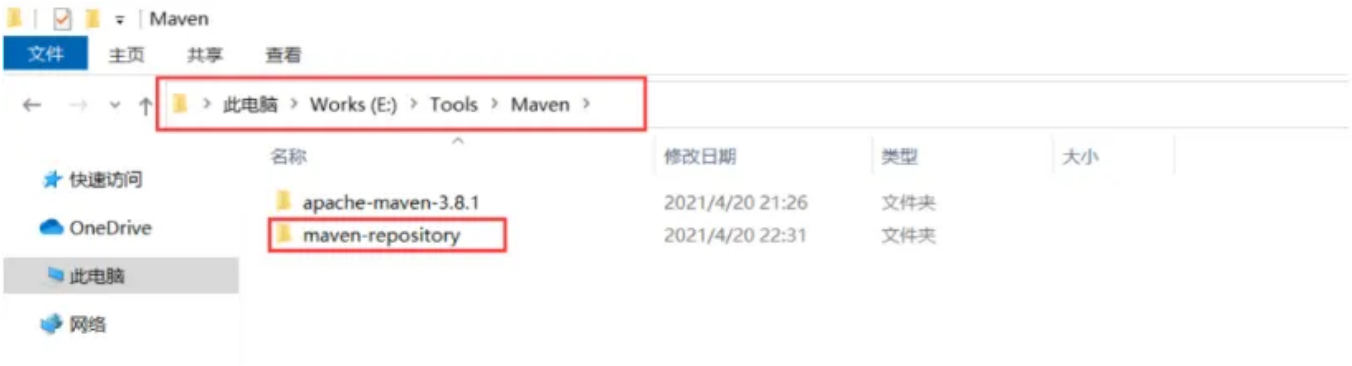


4.然后win+R运行cmd，输入mvn –version，如图所示则配置成功

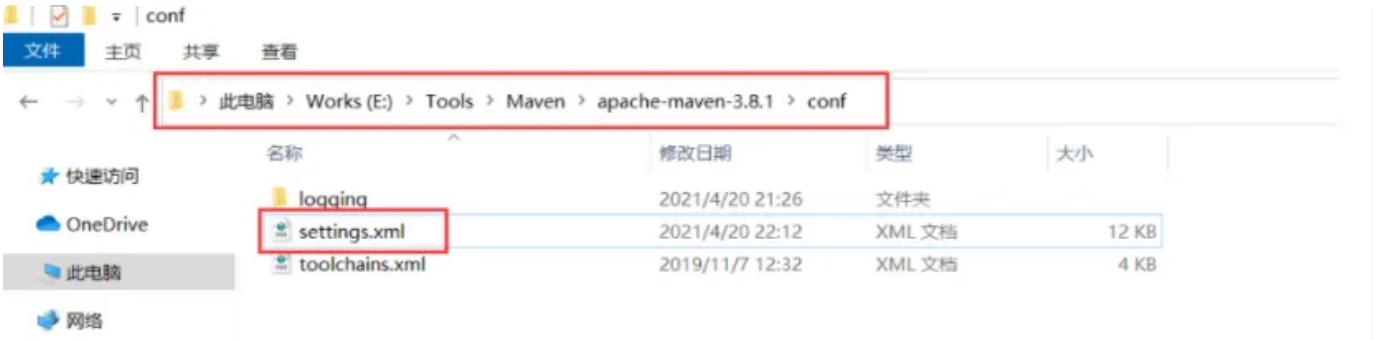


三、配置本地仓库

1.在E:\Tools\Maven\路径下新建maven-repository文件夹，用作maven的本地库。

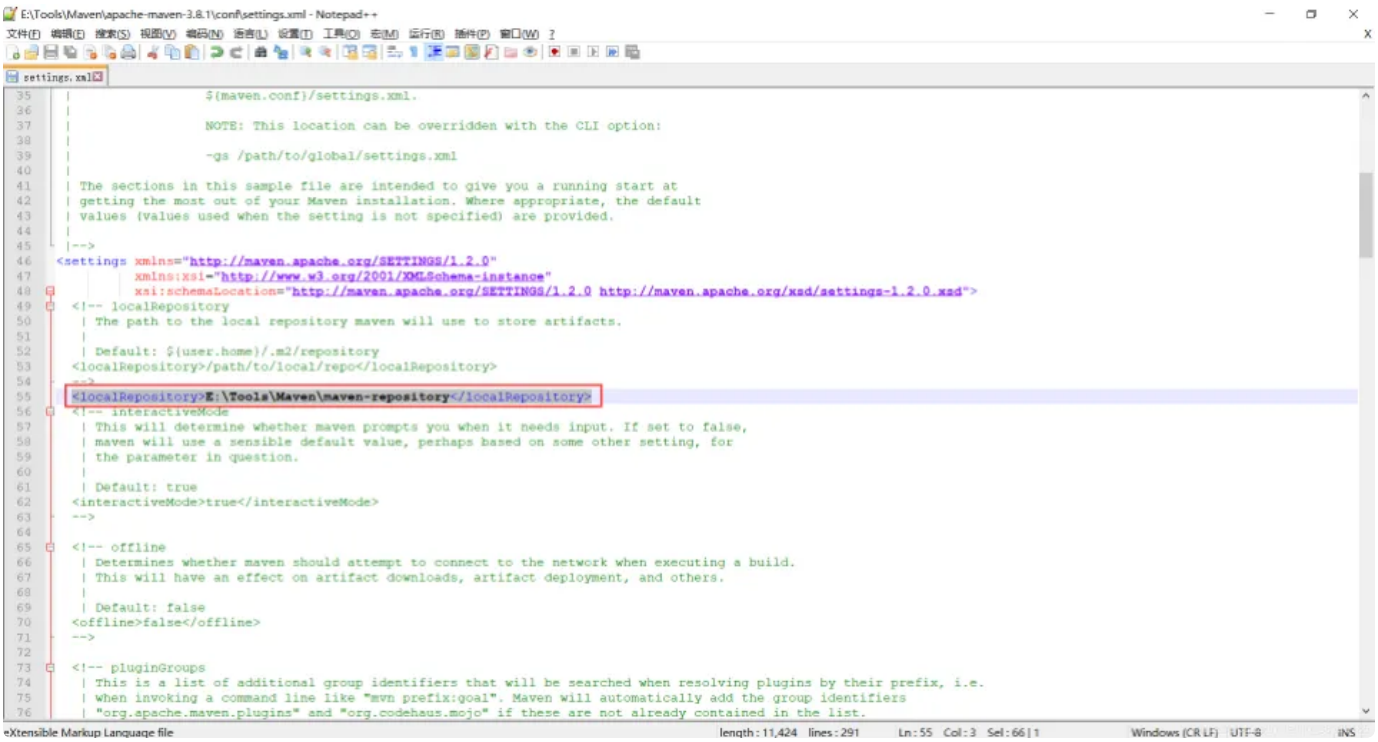


2.在路径E:\Tools\Maven\apache-maven-3.8.1\conf下找到settings.xml文件



3.找到节点localRepository，在注释外添加

```Java
<localRepository>E:\Tools\Maven\maven-repository</localRepository>
```

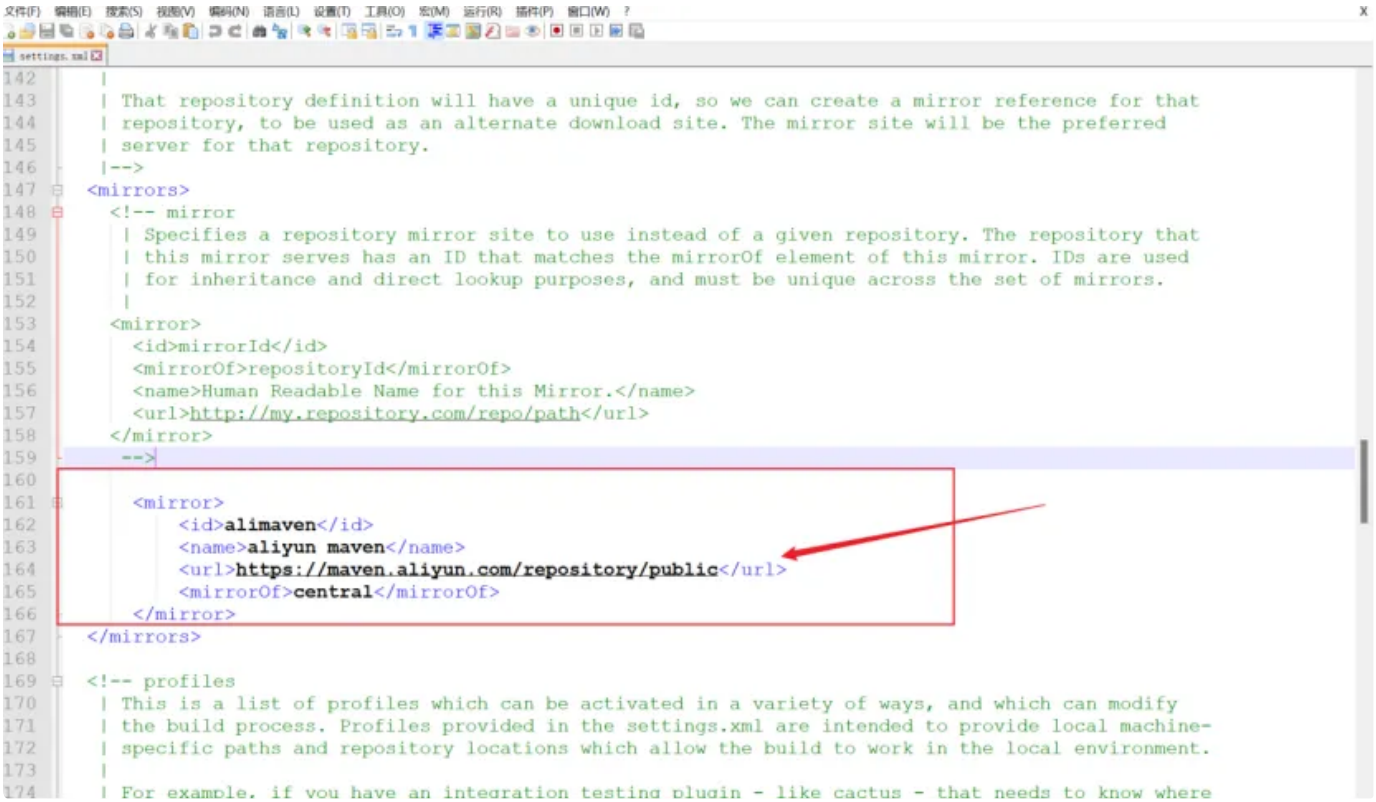


四、配置镜像

1.在settings.xml配置文件中找到mirrors节点

2.添加如下配置（注意要添加在<mirrors>和</mirrors>两个标签之间，其它配置同理）

```java
<!-- 阿里云仓库 -->
<mirror>
    <id>alimaven</id>
    <name>aliyun maven</name>
    <url>http://maven.aliyun.com/repository/public</url>
    <mirrorOf>central</mirrorOf>
</mirror>
```



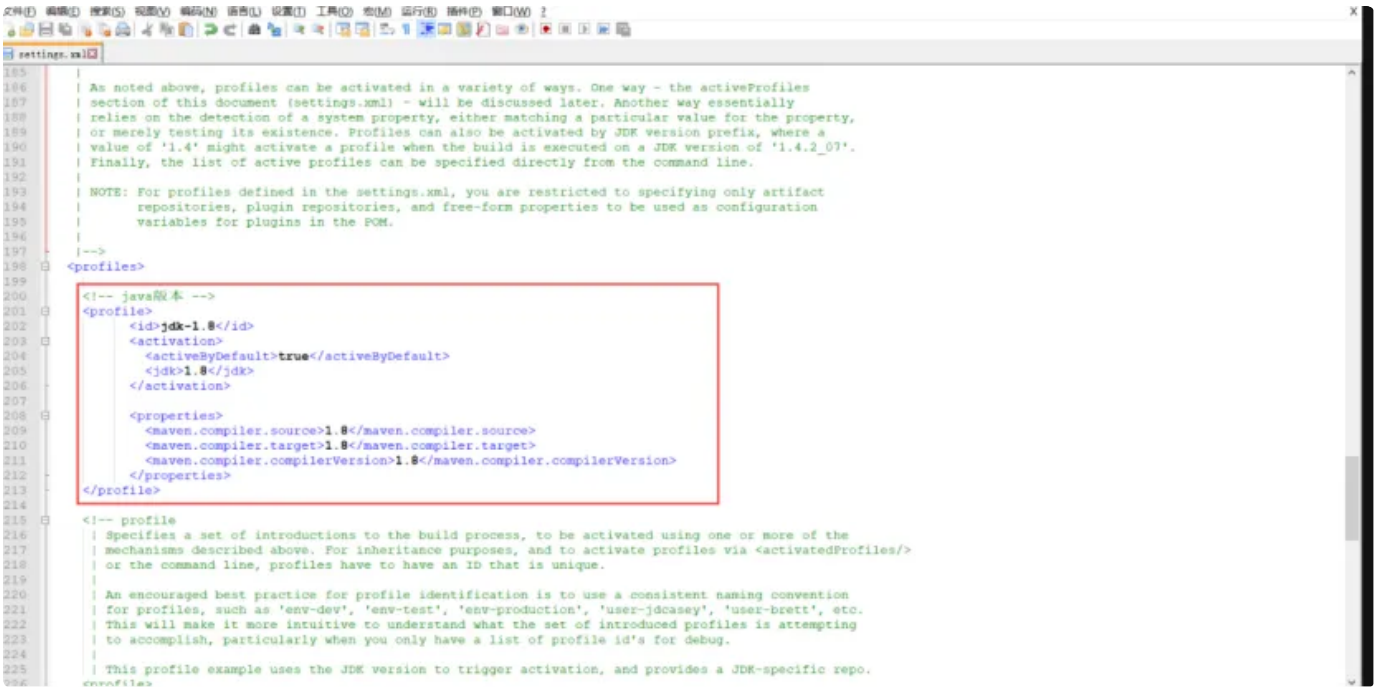
五、配置JDK

1. 在settings.xml配置文件中找到profiles节点

2. 添加如下配置

```java
<profile>
      <id>jdk-1.8</id>
      <activation>
        <activeByDefault>true</activeByDefault>
        <jdk>1.8</jdk>
      </activation>

      <properties>
        <maven.compiler.source>1.8</maven.compiler.source>
        <maven.compiler.target>1.8</maven.compiler.target>
        <maven.compiler.compilerVersion>1.8</maven.compiler.compilerVersion>
      </properties>
</profile>
```



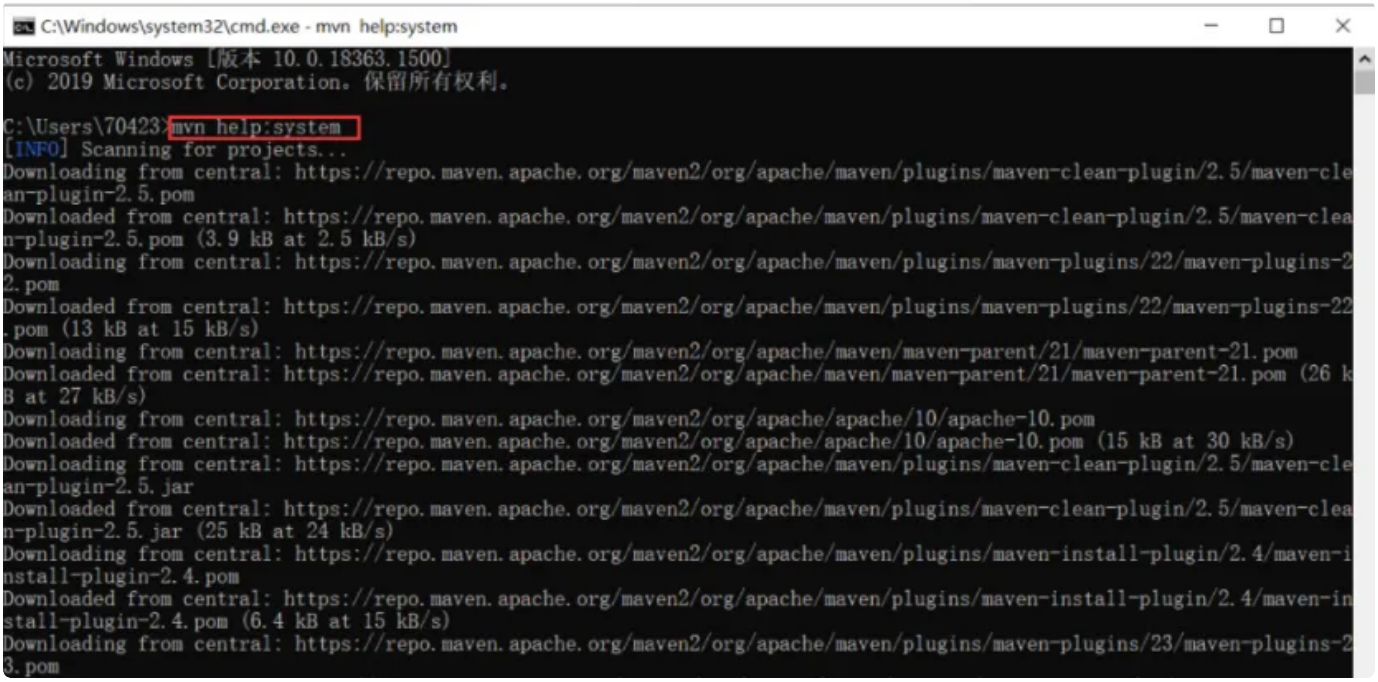
```
  |
  | As noted above, profiles can be activated in a variety of ways. One way - the activeProfiles
  | section of this document (settings.xml) - will be discussed later. Another way essentially
  | relies on the detection of a system property, either matching a particular value for the property,
  | or merely testing its existence. Profiles can also be activated by JDK version prefix, where a
  | value of '1.4' might activate a profile when the build is executed on a JDK version of '1.4.2_07'.
  | Finally, the list of active profiles can be specified directly from the command line.
  |
  | NOTE: For profiles defined in the settings.xml, you are restricted to specifying only artifact
  |       repositories, plugin repositories, and free-form properties to be used as configuration
  |       variables for plugins in the POM.
  |
  |-->
<profiles>

<!-- java版本 -->
<profile>
      <id>jdk-1.8</id>
      <activation>
        <activeByDefault>true</activeByDefault>
        <jdk>1.8</jdk>
      </activation>

      <properties>
        <maven.compiler.source>1.8</maven.compiler.source>
        <maven.compiler.target>1.8</maven.compiler.target>
        <maven.compiler.compilerVersion>1.8</maven.compiler.compilerVersion>
      </properties>
</profile>

  <!-- profile
   | Specifies a set of introductions to the build process, to be activated using one or more of the
   | mechanisms described above. For inheritance purposes, and to activate profiles via <activatedProfiles/>
   | or the command line, profiles have to have an ID that is unique.
   |
   | An encouraged best practice for profile identification is to use a consistent naming convention
   | for profiles, such as 'env-dev', 'env-test', 'env-production', 'user-jdcasey', 'user-brett', etc.
   | This will make it more intuitive to understand what the set of introduced profiles is attempting
   | to accomplish, particularly when you only have a list of profile id's for debug.
   |
   | This profile example uses the JDK version to trigger activation, and provides a JDK-specific repo.
```

3.配置完成，win+R运行cmd，输入mvn help:system测试，配置成功则本地仓库（E:\Tools\Maven\maven-repository）中会出现一些文件



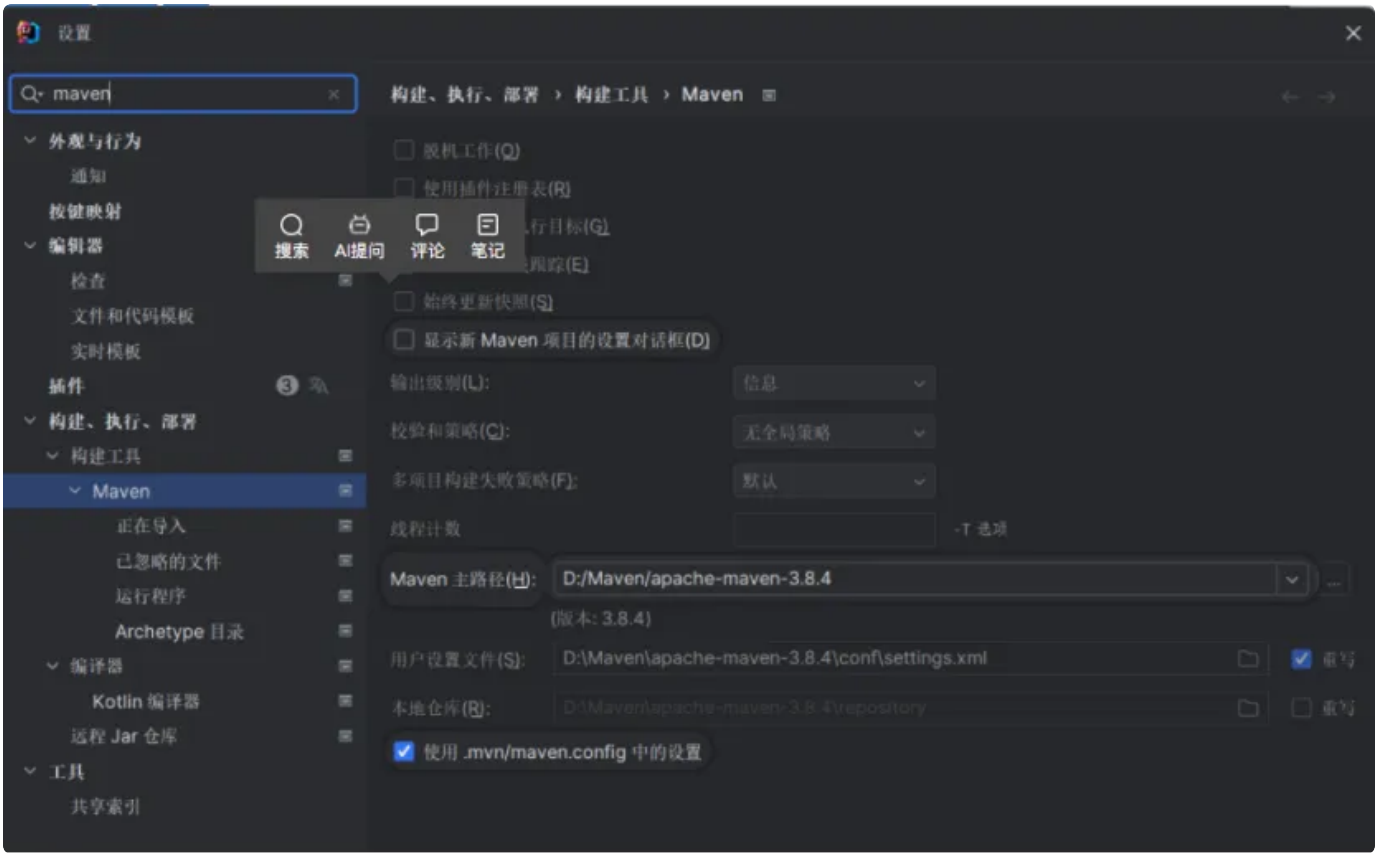
```
C:\Windows\system32\cmd.exe - mvn help:system
Microsoft Windows [版本 10.0.18363.1500]
(c) 2019 Microsoft Corporation。保留所有权利。

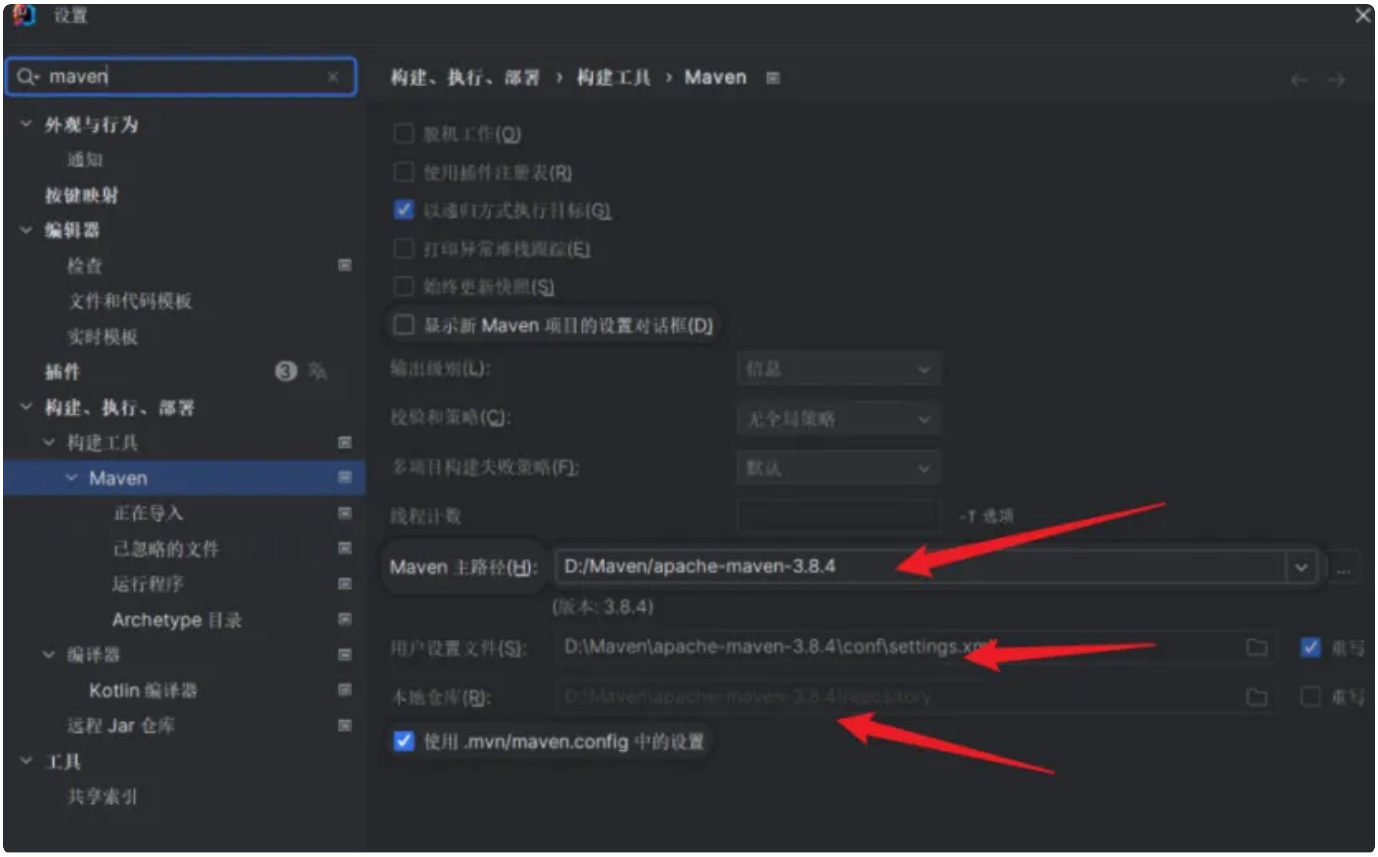
C:\Users\70423>mvn help:system
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-cle
an-plugin-2.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clea
n-plugin-2.5.pom (3.9 kB at 2.5 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/22/maven-plugins-2
2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/22/maven-plugins-22
.pom (13 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/21/maven-parent-21.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/21/maven-parent-21.pom (26 k
B at 27 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/10/apache-10.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/10/apache-10.pom (15 kB at 30 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-cle
an-plugin-2.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/2.5/maven-clea
n-plugin-2.5.jar (25 kB at 24 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-i
nstall-plugin-2.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/2.4/maven-in
stall-plugin-2.4.pom (6.4 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/23/maven-plugins-2
3.pom
```

六、添加到IDEA

1.注意设置到 IDEA 的全局中


欢迎访问 IntelliJ IDEA
IntelliJ IDEA
2023.2.3
项目
远程开发
Beta
SSH
JetBrains Space
WSL
自定义
插件
学习
颜色方案
Dark
与操作系统同步
无障碍功能
IDE 字体:
13.0
针对红-绿色觉缺陷调整颜色
运作方式
按键映射
Windows
配置...
导入设置...
所有设置...


设置
maven
外观与行为
通知
按键映射
编辑器
检查
文件和代码模板
实时模板
插件
构建、执行、部署
构建工具
Maven
正在导入
已忽略的文件
运行程序
Archetype 目录
编译器
Kotlin 编译器
远程 Jar 仓库
工具
共享索引
搜索
AI提问
评论
笔记
构建、执行、部署 › 构建工具 › Maven
脱机工作(O)
使用插件注册表(R)
始终更新快照(S)
显示新 Maven 项目的设置对话框(D)
输出级别(L):
信息
校验和策略(C):
不全局策略
多项目构建失败策略(F):
默认
线程计数
-T 选项
Maven 主路径(H):
D:/Maven/apache-maven-3.8.4
(版本: 3.8.4)
用户设置文件(S):
D:\Maven\apache-maven-3.8.4\conf\settings.xml
重写
本地仓库(R):
重写
使用 .mvn/maven.config 中的设置



i： Maven主路径设置为自己的安装路径 我这里是: E:\Tools\Maven\apache-maven-3.8.1

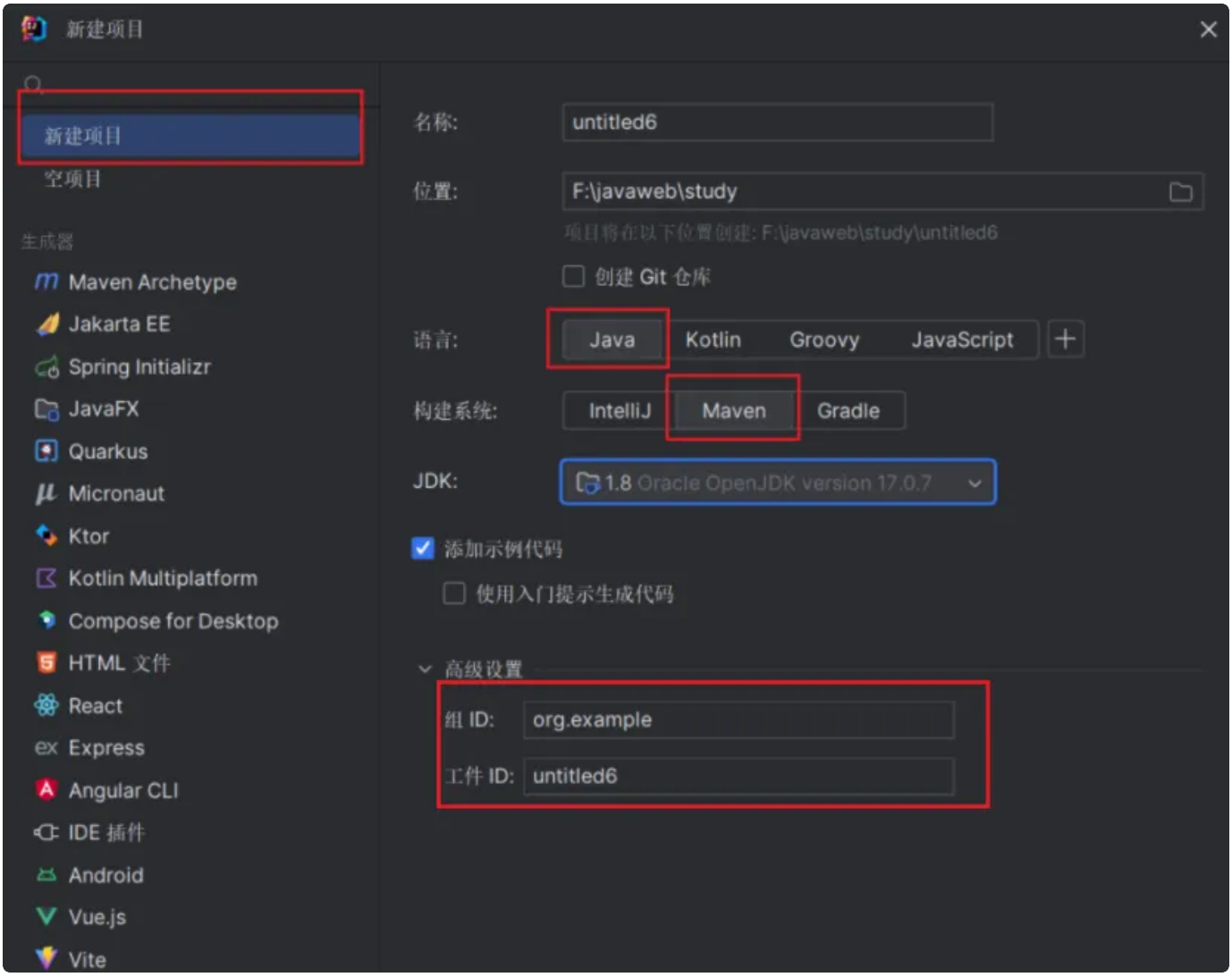
ii： 用户设置文件设置为Maven的conf文件夹下的settings.xml 文件，并注意后面打上对号

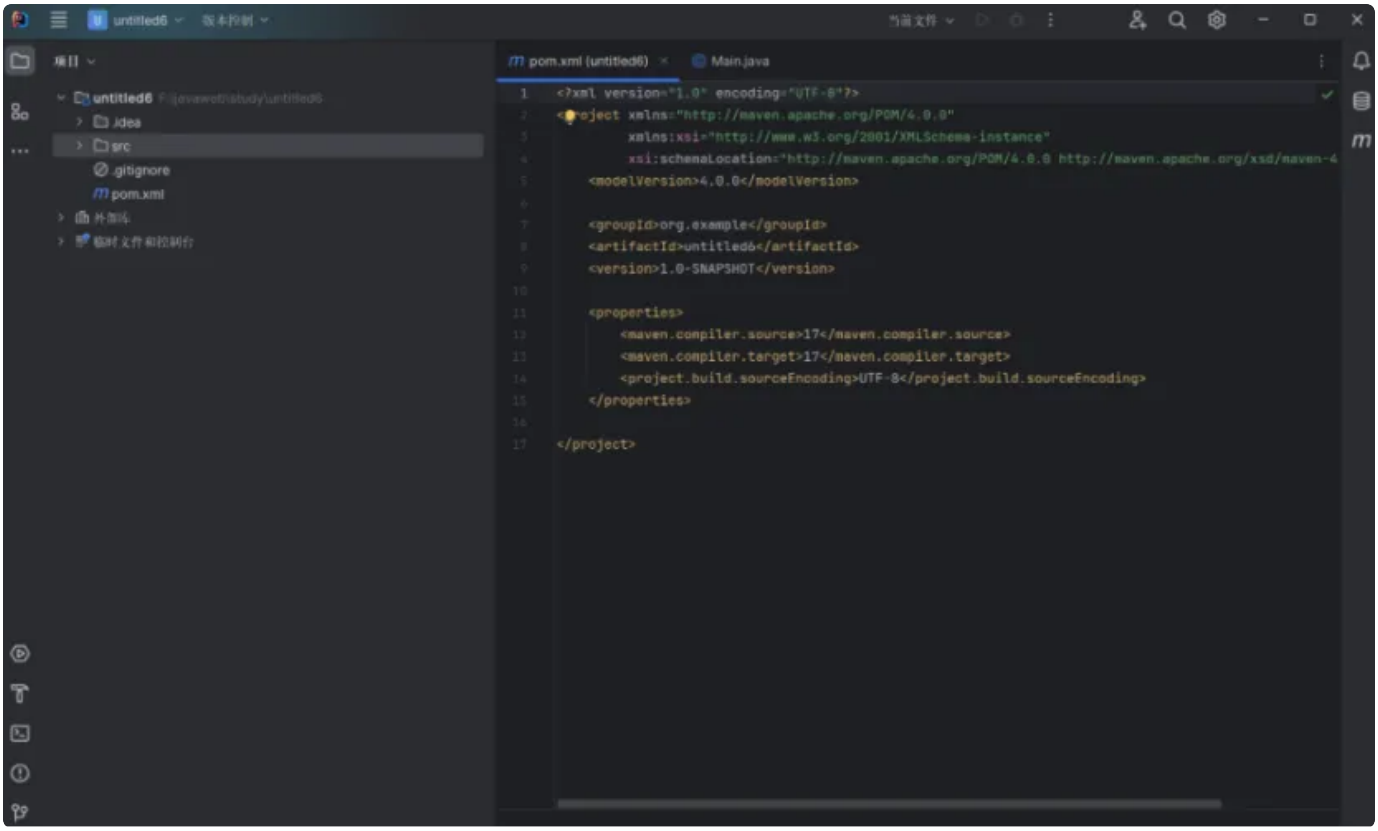
我这里是: E:\Tools\Maven\apache-maven-3.8.1\conf\settings.xml

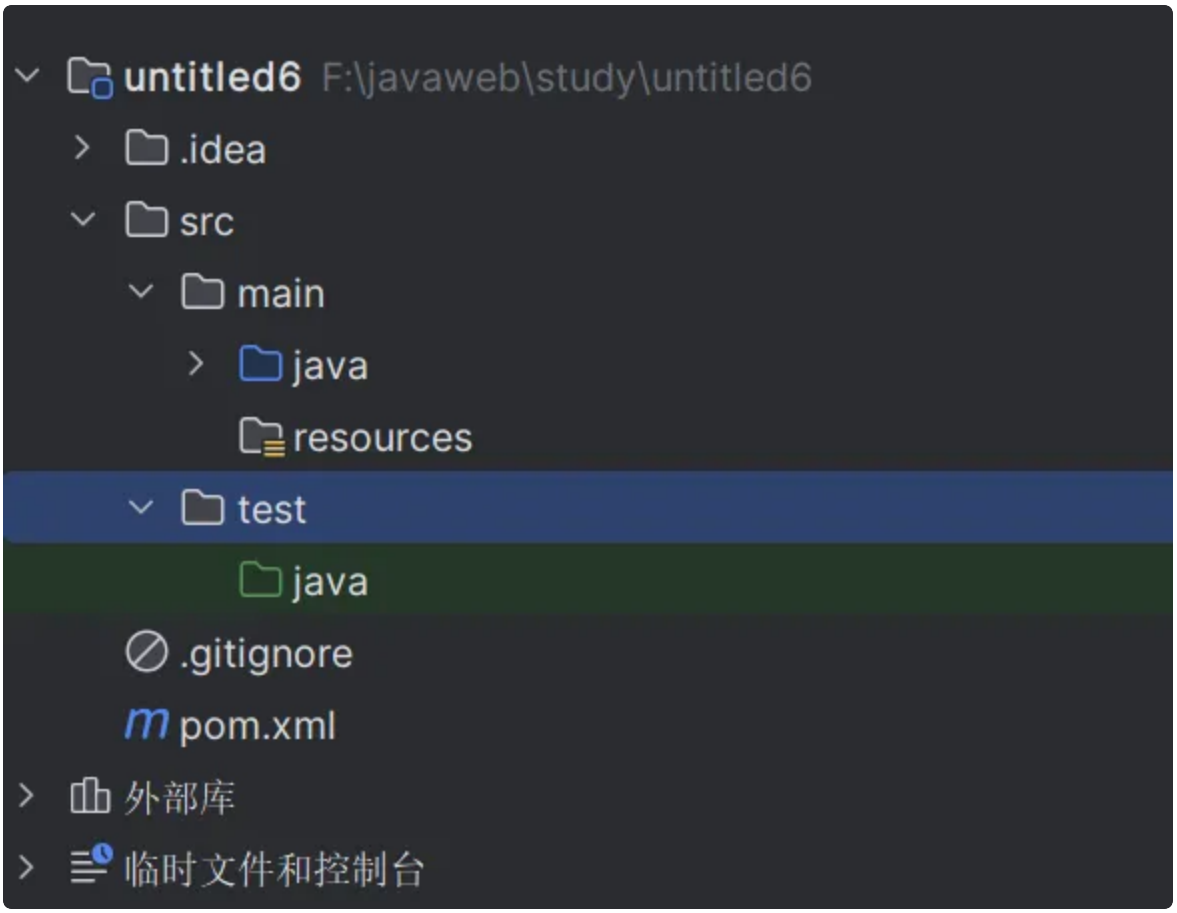
iii： 本地仓库注意看看是否是自己设置的仓库，不是的话，修改下即可。

我这里是: E:\Tools\Maven\apache-maven-3.8.1\maven-repository

IDEA创建Maven项目


新建项目
新建项目
空项目
生成器
Maven Archetype
Jakarta EE
Spring Initializr
JavaFX
Quarkus
Micronaut
Ktor
Kotlin Multiplatform
Compose for Desktop
HTML 文件
React
Express
Angular CLI
IDE 插件
Android
Vue.js
Vite
名称:
untitled6
位置:
F:\javaweb\study
项目将在以下位置创建: F:\javaweb\study\untitled6
创建 Git 仓库
语言:
Java
Kotlin
Groovy
JavaScript
构建系统:
IntelliJ
Maven
Gradle
JDK:
1.8 Oracle OpenJDK version 17.0.7
添加示例代码
使用入门提示生成代码
高级设置
组 ID:
org.example
工件 ID:
untitled6


untitled6
pom.xml (untitled6)
Main.java
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4
<modelVersion>4.0.0</modelVersion>
<groupId>org.example</groupId>
<artifactId>untitled6</artifactId>
<version>1.0-SNAPSHOT</version>
<properties>
<maven.compiler.source>17</maven.compiler.source>
<maven.compiler.target>17</maven.compiler.target>
<project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
</properties>
</project>


untitled6 F:\javaweb\study\untitled6
.idea
src
main
java
resources
test
java
.gitignore
pom.xml
外部库
临时文件和控制台

配置pom.xml文件

```Java
<dependencies>
    <dependency>
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-java</artifactId>
        <version>4.0.0</version>
    </dependency>
    <dependency>
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-api</artifactId>
        <version>4.0.0</version>
    </dependency>

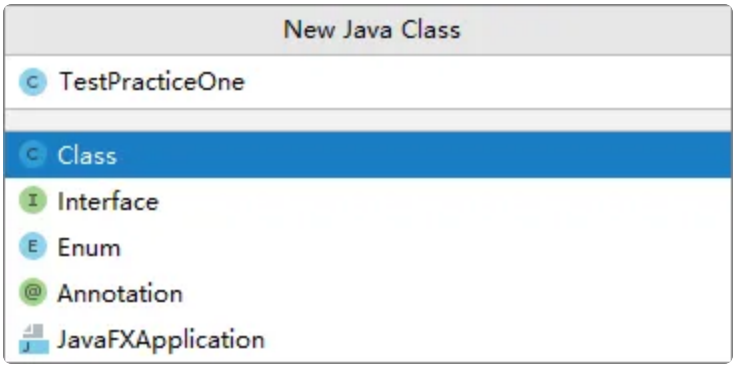
    <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter-api</artifactId>
        <version>5.0.0</version>
        <scope>test</scope>
    </dependency>

    <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter-engine</artifactId>
        <version>5.0.0</version>
        <scope>test</scope>
    </dependency>

    <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter-params</artifactId>
        <version>5.0.1</version>
        <scope>test</scope>
    </dependency>
    <!-- Apache Commons IO -->
    <dependency>
        <groupId>commons-io</groupId>
        <artifactId>commons-io</artifactId>
        <version>2.11.0</version>
    </dependency>

</dependencies>
```

编写Selenium测试代码


New Java Class
TestPracticeOne
Class
Interface
Enum
Annotation
JavaFXApplication

```java
import org.apache.commons.io.FileUtils;
import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.openqa.selenium.OutputType;
import org.openqa.selenium.TakesScreenshot;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import java.io.File;
import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.concurrent.TimeUnit;


public class TestPracticeOne {
    private WebDriver driver;
    @BeforeEach
    public void setup() {
        //提交最终代码脚本时，请将驱动路径换回官方路径"C:\\Users\\86153\\AppData\\Local\\Google\\Chrome\\Application\\chromedriver.exe"
        System.setProperty("webdriver.chrome.driver", "C:\\Users\\86153\\AppData\\Local\\Google\\Chrome\\Application\\chromedriver.exe");
        ChromeOptions chromeOptions = new ChromeOptions();
        chromeOptions.addArguments("--remote-allow-origins=*");
        driver = new ChromeDriver(chromeOptions);
        driver.get("https://www.ctrip.com");
        driver.manage().window().maximize();
        driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);

    }
    // test-code-start

    // 请在此处插入Selenium脚本代码

    // test-code-end


    @AfterEach
    public void teardown() {
        this.driver.quit();
    }

    private void takeScreenshot(String fileName) {
        SimpleDateFormat dateFormat = new SimpleDateFormat("HHmmssddSSS");
        String timestamp = dateFormat.format(new Date());
```

```
44          String timestampedFileName = timestamp + "_" + fileName;
45          File screenshotsDir = new File("screenshots");
46          if (!screenshotsDir.exists()) {
47              screenshotsDir.mkdirs();
48          }
49          String screenshotFilePath = screenshotsDir.getAbsolutePath() + Fil
    e.separator + timestampedFileName;
50          File screenshotFile = ((TakesScreenshot) driver).getScreenshotAs(O
    utputType.FILE);
51          try {
52              FileUtils.copyFile(screenshotFile, new File(screenshotFilePath
    ));
53          } catch (Exception e) {
54              e.printStackTrace();
55          }
56      }
57  }
58
```